Requirements Gathering: MobiGuide (Market Driven Phone Recommender)

1. Identify Stakeholders

- **Internal Stakeholders:** DEPI Project Instructors
- **External Stakeholders (End Users):** Egyptian Mobile Consumers, E-Commerce Buyers looking for specific device specifications within budget constraints (Gamers, Content Creators, Everyday Users).

2. Plan the Approach

- **Objective:** Aggregate smartphone listings across Amazon, Noon, and Jumia to eliminate market confusion, enable cross-platform price matching, and deliver automated persona-based purchase recommendations.
- **Timeline:** 4-Week Sprint (Week 1: Data Gathering & Scraping → Week 2: ETL & DAX Modeling → Week 3: Dashboard Development → Week 4: QA, Testing & Presentation).
- **Elicitation Techniques:** Simulated User Surveys (Persona Mapping), Competitor Analysis (Price Comparison Platforms), and Mentorship Feedback Workshops.

3. Conduct Elicitation (Core Business Needs)

- **Customer Need:** Ability to quickly find the best mobile phone suited for specific usage (e.g., Gaming/Battery vs. Camera/Daily Use) without manually checking multiple websites.
- **Price Transparency Need:** Ability to compare real-time prices for the same device across Amazon Egypt, Noon, and Jumia to pinpoint the cheapest available deal.

4. Document Observations and Findings

- **Functional Requirements (What the system must do):**
    - Automatically scrape listing details (Brand, Model, Price, RAM, Storage, Battery, Refresh Rate, Processor, Camera MP, Rating) from 3 e-commerce platforms.
    - Categorize all devices into 4 price tiers: Budget (<10k EGP), Midrange (10k–25k EGP), Upper Midrange (25k–45k EGP), and Flagship (>45k EGP).
    - Calculate custom performance scores: Gaming_Score, Camera_Score, Battery_Score, and Value_for_Money_Score.

    - Dynamically highlight the lowest price among Amazon, Noon, and Jumia for each phone model.

- **Non-Functional Requirements (How it must perform):**
    - Dashboard interaction response time under 2 seconds upon filter selection.
    - Clean UI/UX with consistent theme, smooth side navigation, and mobile-friendly visual sizing.
- **Constraints:**
    - Dependency on public e-commerce layout stability for web scraping.
    - Currency & price volatility in the Egyptian smartphone market.

5. Analyze and Prioritize (MoSCoW Framework)

- **Must Have:** Data scraping pipeline (Amazon, Noon, Jumia), 4-Tier price classification, basic DAX scoring logic, and Persona 1 & Persona 2 recommendation dashboards.
- **Should Have:** Cross-platform price matcher (Cheapest Platform Flag), dynamic budget sliders, and interactive brand comparison visuals.
- **Could Have:** Historical price trend tracking and seller rating diversity breakdown.
- **Won't Have (For MVP):** Automated real-time transactional checkout links or live automated daily scraping refresh.

6. Verify and Validate

- **Review Process:** Walkthrough of Power BI wireframes and sample dataset with DEPI mentors to confirm metric logic and visual presentation.
- **Validation:** Testing persona recommendations against real-world tech reviews to ensure output accuracy and scoring reliability.

7. Manage Changes

- **Traceability & Version Control:** Version control for Python scrapers via GitHub and tracked .pbix file backups for Power BI dataset updates.
- **Change Control Process:** Weekly team syncs to evaluate scope changes, algorithm tweaks, or missing dataset attributes before locking final deliverables.